



Federal Office
for Information Security

5G Risk Analysis

Detailed Analysis: **Infrastructure**



Version History

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1 List of Abbreviations

3GPP	3rd Generation Partnership Project
5G-GUTI	5G Global Unique Temporary Identifier
BSS	Business Support System
CCM	Container Infrastructure Cluster Management
CIR	Container Image Registry
CIS	Container Infrastructure Service
CISM	Container Infrastructure Service Management
CPU	Central Processing Unit
EM	Element Management
FCAPS	Fault, Configuration, Accounting, Performance, Security
GPU	Graphics Processing Unit
HSM	Hardware Security Module
IMEI	International Mobile Equipment Identity
IMS	IP Multimedia Subsystem
MANO	Management and Orchestration
MEC	Multi Access Edge Computing
MNO	Mobile Network Operator
NF	Network Function
NFV	Network Function Virtualization
NFVI	Network Function Virtualization Infrastructure
NFVO	Network Function Virtualization Orchestrator
NS	Network Service
OSS	Operational Support System
RAN	Radio Access Network
SDN	Software-Defined Networking
SUPI	Subscription Permanent Identifier
TLS	Transport Layer Security
UDM/HSS	Unified Data Management / Home Subscriber Server
UPF	User Plane Function
VIM	Virtual Infrastructure Manager
VM	Virtual Machine
VNF	Virtual Network Functions
VNFC	Virtualized Network Function Components

2 Introduction and Purpose

2.1 Intention

The intention of the document is to dive deep into the analysis of risk scenarios where the **Infrastructure** layer is the attacked or failing subsystem of a 5G network.

This document shows the current view on the probability that any risk scenarios could become reality.

The study is only available in English language, as the target group is primarily the international IT security community. The study is also in the context of English publications on this topic.

2.2 Link to Framework Document

This document is linked to the framework document and provides relevant information about the Infrastructure layer in 5G networks and a probability assessment of relevant risk scenarios described in the framework document. The overall risk assessment for each risk scenario consists of the results of this analysis and the impact described in the framework document. The overall risk assessment for the Infrastructure layer is summarized in Chapter 6.2 of this document and in the framework document.

3 Methodology

The methodology used for the 5G Risk Analysis (5G-RA) is based on PASTA. In this document and the related documents, we will refer to the methodology as 5G-RA.

This chapter explains the seven steps of the 5G-RA methodology and assigns them to the corresponding PASTA steps.

Step 1: Definition of Risk Scenarios

In the first step of the 5G-TARA, the 5G network needs to be analysed from the following three perspectives:

1. Public Mobile Network Operator (MNO) Perspective,
2. Private 5G Network Operator Perspective,
3. User Perspective.

Several risks with a significant impact on the society or public safety have been identified by the legislative in the past. To mitigate these risks, laws were set into force, e.g. the legislative to ensure emergency calls are available (TKG §165). Most of these laws define fines for mobile network operators if the regulative obligation is not followed.

These perspectives are explained in detail in Chapter 6 of the framework document. The goal is to identify the possible risks that the 5G network might be exposed to from each perspective.

Step 2: 5G System Diagram

In step 2, a detailed system diagram of the parts of a 5G network needs to be created. This diagram will serve as a basis for the following steps.

Step 3: Refinement of Detailed 5G Subsystem Diagrams

In this step, more technical definitions and explanations of the relevant 5G subsystems are introduced. For example, various network functions are part of the 5G core network. This level of detail serves as a basis for the following steps.

Step 4: Collection of Known 5G Threats

In this step, publicly known 5G specific threats are collected. For this purpose, relevant studies and input documents are collected. This is the basis for the identification of threats in the next step.

Step 5: Search of 5G Vulnerabilities

Step 5 includes the analysis and identification of relevant threats for each risk scenario. The findings of this step are part of the corresponding Detailed Risk Analyses.

Step 6: Examination of Identified Threats

In this step, all threats that were found in step 5 are examined on whether there is a possibility for this threat to be realized. Hypothetical scenarios are used as well as references to successful attacks, which were performed in the past.

Step 7: Risk Calculation

The last step of the methodology is a calculation of the risk for all threats that were identified in step 5 and are still considered relevant after step 6 was carried out. The overall risk assessment is presented in Chapter 6 of this document.

The following table assigns the steps of the 5G-RA to the steps of PASTA and gives an overview of the document containing the results of each step.

Table 1: Assignment of 5G-RA steps to PASTA-steps

#	5G-RA methodology	Document Layer	PASTA
1	Definition of Risk Scenarios	Framework	Define Objectives
2	5G System Diagram	Framework	Define Technical Scope
3	Refinement of Detailed 5G Subsystem Diagrams	Framework & Detailed Risk Analysis	Application Decomposition
4	Collection of Known 5G Threats	Detailed Risk Analysis	Threat Analysis
5	Search of 5G Vulnerabilities	Detailed Risk Analysis	Vulnerability and Weakness Analysis
6	Examination of the Identified Threats	Detailed Risk Analysis	Attack Modelling
7	Risk Calculation	Framework & Detailed Risk Analysis	Risk and Impact Analysis

4 5G Network Model: Infrastructure

The 5G network model introduced in the framework document is used to split the 5G risk analysis into multiple Detailed Risk Analyses. The network model follows standards (like 3GPP standards) as much as possible and comprises a variety of interconnected subsystems that are categorized on a high-level view as shown in Figure 1. 5G networks may consist of any combination of 5G Standalone and 5G Non-Standalone architecture options, which therefore includes also 4G RAN (E-UTRAN) and 4G-CORE (EPC). However, this risk analysis covers only 5GC for the core. All other subsystems are not limited to 5G.

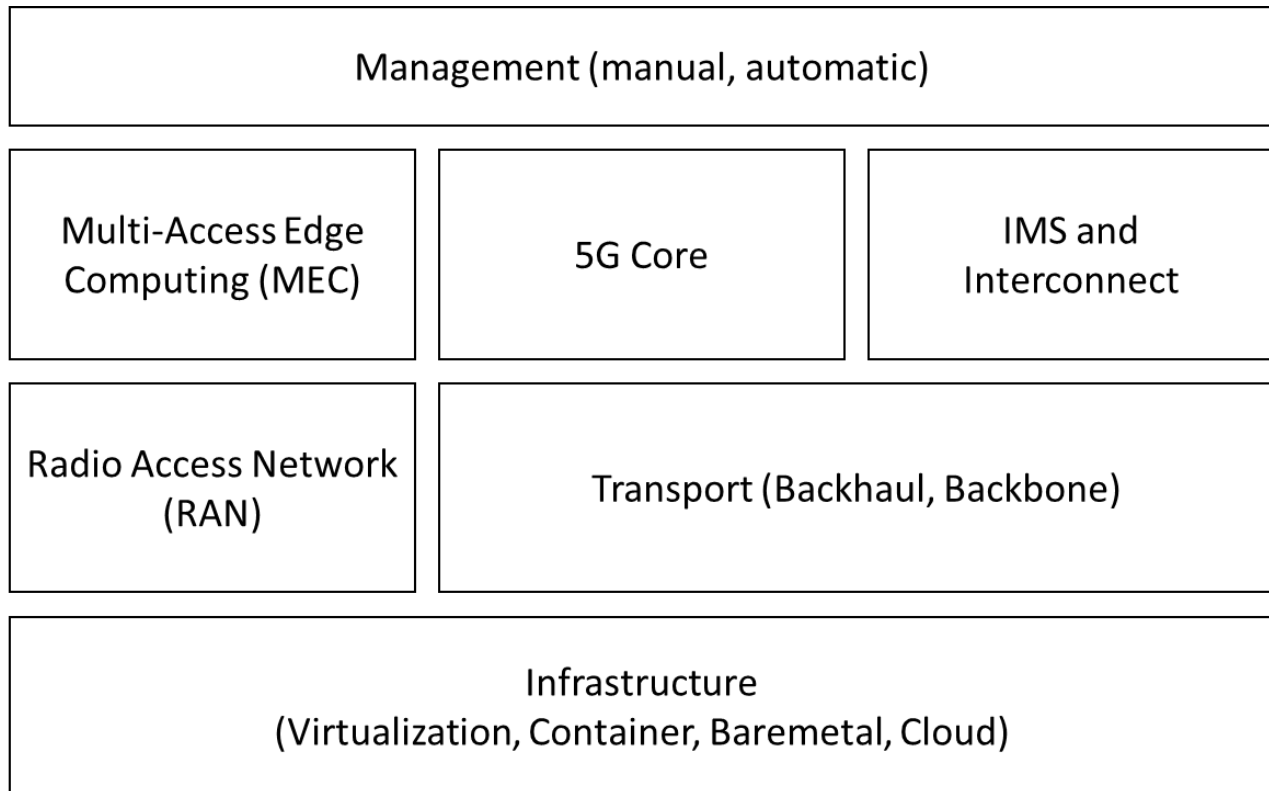


Figure 1: 5G Network Model: High level view

Other subsystems are not specifically designed for the 5G architecture and used either to operate the 5G system or to connect it to other internal and external networks.

The 5G technical specifications are fundamentally infrastructure agnostic. That is, they do not prescribe a certain implementation of or make assumptions about the underlying infrastructure. 5G mobile network deployments may consist of physical network functions, virtualized network functions, containerized network functions, or a mix of all of these. Therefore, the ETSI Network Function Virtualization (NFV) framework is taken as a basis for performing the following 5G infrastructure risk assessment, as it provides a consolidated view of all these different deployments options. Particular attention is given to the Network Function Virtualization Infrastructure (NFVI), the virtualized and containerized Virtual Network Functions (VNF) deployed on top of it as well as the associated Management and Orchestration (MANO) functions.

5G Infrastructure System Diagram

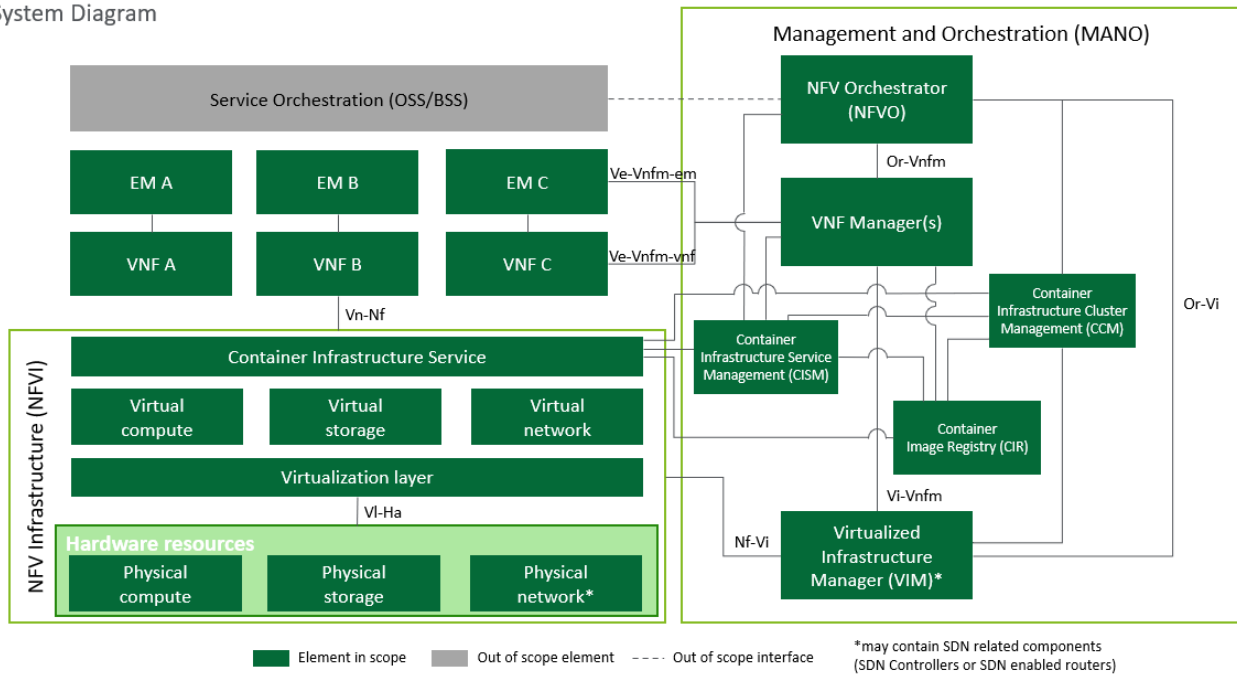


Figure 2: 5G Infrastructure – System Diagram

4.1 NFV Infrastructure (NFVI)

Hardware resources

The hardware resources provide the underlying physical infrastructure for the NFV environment. This includes compute resources (e.g., central processing units, graphics processors), storage (e.g., main memory, persistent storage), and networking equipment (e.g., routers, switches, cables). Depending on the type of deployment, this network equipment may include physical switches featuring support for software-defined networking (SDN), controlled by a SDN controller which may be located, for example, in the Management and Orchestration domain. (See also [1])

Virtualization layer and virtual resources

The virtualization layer abstracts virtual resources away from the physical infrastructure. As such, it is responsible for (see [1]):

- “Abstracting and logically partitioning physical resources, commonly as a hardware abstraction layer.
- Enabling the software that implements the VNF to use the underlying virtualised infrastructure.
- Providing virtualised resources to the VNF, so that the latter can be executed.”

The technical components that are commonly found in the virtualization layer are hypervisors, storage virtualization software, and SDN Data Plane applications.

Container Infrastructure Service

Depending on the type of deployment, mobile network operators may choose to deploy their virtual network functions in a containerized fashion, rather than the more traditional virtual machines. In this case, the NFVI also contains a Container Infrastructure Service, also called container engine, that allows for the execution of containerized applications. Importantly, the CIS itself may run on top of virtualized nodes or on top of bare-metal nodes (i.e., physical servers) directly. (See also [2]).

4.2 Virtual Network Functions and Element Management

VNFs are virtualized implementations of mobile network functions (e.g., an AMF). They are comprised of one or more Virtualized Network Function Components (VNFC) each providing a subset of the VNF's functionality. For example, one VNFC may run a database, another may expose a webserver.

The element management (EM) performs a variety of management functions for one or more VNFs. This may include multiple applications performing the essential management functions commonly abbreviated FCAPS (fault, configuration, accounting, performance, security).

4.3 Management and Orchestration (MANO)

NFV Orchestrator

The NFV Orchestrator's main responsibility is Network Service (NS) management, i.e., managing the overall deployment of multiple connected VNFs. In this capacity, it also performs the high-level coordination of MANO components, including VNF Manager(s) and the Virtualized Infrastructure Manager. Further, it plays an important role in resource management and capacity management as well as exposing various management services towards other Support Systems (OSS/BSS). (See also [3]).

VNF Manager

VNF Managers are responsible for lifecycle management of individual VNF instances, including instantiation, update, query, scaling, and termination. They may serve one or multiple VNFs connecting to both the VNF instances and the associated Element Management components. In this role, they also fulfill important monitoring and event reporting functions. (See also [3]).

Virtualized Infrastructure Manager

The Virtualized Infrastructure Manager controls NFVI compute, storage, and network resources. In addition to FCAPS, this includes capacity management, resource reservation and quota management as well as service exposure and notification towards VNFM and NFVO. (See also [3]).

Container Infrastructure Cluster Management

The CCM is responsible for controlling and managing CIS clusters (e.g., Kubernetes clusters). Beyond that, it also performs FCAPS management tasks at a cluster level. (See also [2]).

Container Infrastructure Service Management

The CISM has two main responsibilities: workload management and node management. The former includes services, such as lifecycle management for containerized workloads, managing namespaces in a container cluster, and managing namespace quotas. The latter comprises tasks such as lifecycle management for container instances and managed cluster objects. (See also [2], [4]).

Container Image Registry

The CIR is primarily responsible for container image management, i.e., storing container images and providing related services to other functions. These services may include adding deleting, or inquiring about

container images in the CIR, notifying other functions about container image changes, and enforcing access control to the stored information. (See also [4]).

4.4 Interfaces and reference points

VI-Ha (Virtualization layer – Hardware resources)

The VI-Ha interface connects the virtualization layer with the underlying hardware. As outlined in [5], a hypervisor would use this interface to pull hardware information and to create virtual resources which are in turn utilized by virtual network functions. (See also [1]).

Vn-Nf (VNF – NFVI)

The Vn-Nf interface connects VNFs and the NFV Infrastructure. As per [1], “it represents the execution environment provided by the NFVI to the VNF” which in turn uses the interface to access the virtual resources.

Nf-Vi (NFVI – VIM)

The NF-Vi interface connects the NFV Infrastructure and the VIM. Information exchanged via this interface includes [1]:

- “Specific assignment of virtualised resources in response to resource allocation requests.
- Forwarding of virtualised resources state information.
- Hardware resource configuration and state information (e.g. events) exchange.”

Vi-Vnfm (VIM – VNFM)

The Vi-Vnfm reference point is located between the VIM and the VNFM. Its primary functionality includes the exchange of information related to [1]:

- “Resource allocation requests by the VNF Manager.
- Virtualised hardware resource configuration and state information (e.g. events) exchange.”

Or-Vi (NFVO – VIM)

The Or-Vi reference point is located between the NFVO and the VIM. Its primary functionality includes [1]:

- “Resource reservation and/or allocation requests by the NFV Orchestrator.
- Virtualised hardware resource configuration and state information (e.g. events) exchange.”

Or-Vnfm (NFVO – VNFM)

The Or-Vnfm reference point is located between the NFVO and the VNFM. Its primary functionality includes [1]:

- “Resource related requests, e.g. authorization, validation, reservation, allocation, by the VNF Manager(s).
- Sending configuration information to the VNF manager, so that the VNF can be configured appropriately to function within the VNF Forwarding Graph in the network service.
- Collecting state information of the VNF necessary for network service lifecycle management.”

Ve-Vnfm (VNF/EM – VNFM)

The Ve-Vnfm reference point is located between VNFM and VNF as well as the associated Element Management functions. Depending on the exact end point, it can be further distinguished into a Ve-Vnfm-vnf and a Ve-Vnfm-em interface. The information exchanged via these interfaces includes [1]:

- “Requests for VNF lifecycle management.
- Exchanging configuration information.
- Exchanging state information necessary for network service lifecycle management.”

4.5 Summary

The infrastructure subsystem is essential for the entire 5G network and the associated subscriber services (hereafter referred to as the *3GPP service layer*), as it provides the foundation for all network functions and elements. Still, there are some components that can be considered more sensitive than others. These are components which have access to and control large areas of the infrastructure, and their compromise or malfunction would likely result in a direct breach of confidentiality, integrity, or availability. This applies specifically to the NFV Orchestrator (NFVO), the VNF Manager(s) and the Virtualized Infrastructure Manager (VIM) as highly relevant for several security aspects. This is largely in line with a previous BSI publication on ‘MANO risk analysis in 5G’ which identifies the *Orchestrator* and *xIM Manager* as having a ‘very high’ damage potential, and the *xNF Manager* as having a ‘high’ damage potential.

5 Risk Scenarios

This chapter provides the current view on a range of risk scenarios. It deals with an estimation of probabilities that each of this risk scenarios could become reality. These estimations are based on detailed discussions of potential scenarios of weaknesses or attacks. These discussions conclude with an assessment of the probability of occurrence.

The following table shows if the Infrastructure subsystem is a relevant subsystem of each risk scenario. The risk scenarios themselves are further described in the framework document.

This risk analysis focuses on technical risks and risk scenarios. Covering political aspects is out of scope. Not considered are risk scenarios related to the question, if the manufacturer of the equipment and the supply chain is trustworthy. On this topic exist various publications for IT systems in general [ENI]. A second risk not covered by this analysis is the not standard-conform implementation of components. This analysis assumes vendors comply with 3GPP and other standards as long as a standard for the functionality exists.

Perspective	Section	Risk Scenario	Infrastructure	CIA
Public MNO Perspective	5.2.1	Emergency Call (TKG § 164)		A
	5.2.2	Cell Broadcast (TKG § 164a)		A
	5.2.3	Telecommunications Secrecy (TTDSG § 3)	x	C
	5.2.4	Emergency Preparedness (TKG § 184)		A
	5.2.5	Regional Network Failure		A
	5.2.6	Multi-Regional Network Failure	x	A
	5.2.7	Failure of Mobile Service		A
	5.2.8	Fraud		I
	5.2.9	Subscriber Data Leak	x	C
User Perspective	5.3.1	Urgent Telephone Calls Not Possible		A
	5.3.2	Authenticity of the Receiver of a Call		I
	5.3.3	Integrity of Transmitted Data	x	I
	5.3.4	User's Identity	x	C
	5.3.5	Eavesdropping of a Communication	x	C
	5.3.6	Slowing Down/Restricting the Connection		A
	5.3.7	Connection Failure		A
	5.3.8	Tapping or Altering of the Geographical Position		C
Private 5G Network Operator Perspective	5.4.1	Network Failure	x	A
	5.4.2	Degraded Services/Connections	x	A
	5.4.3	Information Disclosure		C
	5.4.4	User Identities (Authenticity/ Integrity)		I

5.1 Attack Paths and Risk Sources

For each risk scenario a couple of different attack paths and risk sources will be discussed within this document. The result of the discussion will be a probability level for the impact defined in the respective risk scenario.

An attack path refers to a sequence of steps or actions that an attacker can take to exploit vulnerabilities in a system or network, leading to security issues. It outlines the route or method an attacker follows to achieve his goals, often involving a series of interconnected actions that exploit weaknesses in a system's defences.

A risk source is defined as the origin or cause of a potential risk or threat. A risk source is identified as the factor or element that can lead to vulnerabilities or hazards.

As many of attack paths or sub-sequences of attack paths and risk sources are similar for most of the risk scenarios, each attack path and risk source will be introduced from the perspective of the Infrastructure subsystem in this chapter.

In general, an attacker with legitimate access to MNO's assets that shall be protected is called internal attacker. An internal attacker could have access to physical or cryptographic keys, to buildings or rooms or to networks or systems.

5.1.1 Misconfiguration (AP1)

Misconfiguration as a possible attack path concerns all configurable parts of a system. A legitimate configuration can either be deployed by employees of the MNO or the manufacturer or by configuration tools. The cause of misconfiguration can be either human error thus happening unintentionally, or it can also be intentionally as part of an attack.

Misconfigurations may persist undetected if the impacted functionality was not used regularly or if configuration or security impacts of that functionality were not monitored.

5.1.1.1 Confidentiality (AP1-C)

Ensuring the secure configuration of network components and protocols is essential for keeping sensitive data confidential. Yet, it remains one of the most common issues due to the highly complex and configurable nature of modern IT and infrastructure components [6], threatening both data-in-transit and data-at-rest associated to the 5G System.

Data-in-transit

Insufficient network protection, for example, by means of security protocols such as IPSec and TLS, poses a confidentiality threat to data-in-transit. If this protection is lacking entirely or is insufficient (e.g., due to the use of weak security algorithms), it may allow attackers to eavesdrop on potentially sensitive information exchanged between infrastructure components. These types of attacks can occur at any point of the system, but are especially critical on interfaces between MANO components, such as the NFV Orchestrator, VNF Managers, or VIM Managers, which make extensive use of the TLS protocol (see also [7]). If sensitive management communication (incl. management instructions, configuration data, credentials and secrets) is breached to unauthorized parties, it may allow attackers to cause damage to the deployment at a large scale, affecting not only the MANO itself, but the infrastructure and associated workloads managed by it [8].

Data-at-rest

Misconfigured storage can significantly jeopardize data security in cloud and other virtualized deployments. Instead of storing data in their local file system, virtualized workloads usually connect to centralized storage

resources over the network (e.g., block-level storage or object storage). As such, these storage resources need to be configured separately and hardened in such a way that only legitimate endpoints have access to the stored data. If a tenant fails to do so, it can result in data exposure to unauthorized parties [9]. In the context of the 5G System, this can happen to any Network Function or system that persists data, incl. business support systems, and network management systems. In the worst case, misconfigured storage can expose subscriber identities and other personal information.

A lack of security hardening can also threaten confidentiality in the Infrastructure subsystem. This applies especially to those components that interact directly with the 5G NF workloads, such as the Virtualization layer (i.e., hypervisor, container engine), Element Managers and VNF Managers, as well as other management and monitoring tools. An overly broad attack surface due to the presence of unneeded functionality or access associated to the infrastructure components may allow attackers to elevate their privileges and compromise neighbouring workloads or the virtualization infrastructure itself. (See also [10], [11]).

Additionally, insufficient media sanitization and data deletion practices can leave residual information vulnerable to unauthorized access. This can be particularly damaging in multi-tenant environments, such as a public cloud, but it needs to be considered in other deployment scenarios as well. For example, depending on the deployment, ephemeral container workloads may write data to the local disk of the host machine where it is retained even beyond the container's lifetime.

5.1.1.2 Integrity (AP1-I)

Similar to AP1-C, effective integrity protection requires mobile network operators to appropriately configure their infrastructure components and interfaces, so that attackers cannot successfully tamper with information at lower system layers.

Network integrity

Although the 5G System is supposed to protect User Plane and Control Plane data between Network Functions (e.g., by means of IPsec or TLS), it is recommended to ensure multilayer security by additionally securing data-in-transit at the infrastructure level. On one hand, this avoids single points of failure in case higher system layers do not provide the expected protection for 5G System data. On the other hand, it also protects Control and Management traffic associated to the Infrastructure subsystem itself. Lack of integrity protection may allow attackers to severely jeopardize system operations, for example, by performing an active Man-in-the-middle (MitM) attack, tampering with management instructions or workload images as they are transferred over the network. (See also [12]).

Platform integrity

Beyond data-in-transit, network operators also need to protect the integrity of the infrastructure components hosting 5G NFs. This includes hardening access to the virtualization layer and platform, restricting network access, and applying the latest security patches [13]. Failure to do so may allow malicious actors to meddle with essential software components, such as bootloaders and baseboard firmware, thereby threatening any workloads hosted on top. Due to the nature of these attacks against low-level platform components, they can go unnoticed for a long time while fundamentally undermining the integrity of the entire deployment. (See also [14]).

5.1.1.3 Availability (AP1-A)

Misconfigurations impacting the availability of the 5G System may be caused by different infrastructure components and can vary in their severity – from merely degraded performance to complete unavailability.

Degraded performance

In virtualized deployments, a lack of defined resource limits and insufficient performance isolation between workloads on the same host can negatively impact performance and availability. Such a situation may be intentionally caused by malicious actors controlling workloads in a multi-tenant environment to try and degrade the operation of neighboring workloads. However, it can also occur unintentionally when a certain workload experiences heavy load and temporarily consumes more resources than usual (e.g., a 5G UPF serving a crowded area of the 5G network).

Similarly, failure to restrict the instantiation of new workload instances, also known as *VM Sprawl* or *Container Sprawl*, can also negatively impact other tenants using the same shared platform [15]. Beyond potential effects to the availability of the platform, this also increases the attack surface of the deployment unnecessarily, increasing the likelihood of exposing a weakness allowing malicious actors to compromise the system.

Availability failures

Misconfigured or expired public-key certificates can be the cause for complete failures in communication between network components. If a network operator fails to configure or renew certificates in a timely manner, communication channels relying on them for mutual authentication will fail during connection establishment. Such issues are likely to significantly impact network operations [16].

5.1.2 Insufficient Access Controls (AP2)

Access control involves authentication and authorization, i.e., correctly verifying the identity of a subject (e.g., system users) and assigning them the appropriate permissions. If either of those steps does not function as intended, it may allow unauthorized access by malicious actors potentially resulting in breaches of confidentiality, integrity, or availability failures.

5.1.2.1 Confidentiality (AP2-C)

Insufficient authentication and authorization measures pose security risks for most components in the Infrastructure subsystem. To gain unauthorized access to restricted data and services, attackers may exploit weaknesses, such as flawed authentication methods, weak credentials, or a lack of multi-factor authentication (MFA). Even if MFA is used, legacy technologies such as SMS may allow attackers to compromise these additional authentication credentials [17]. In on-premise deployments the network infrastructure hosting central parts of the 5G System (e.g., 5G Core, IMS) is generally only accessible to internal staff and thus, the attack surface is limited. However, for 5G components, management tools, and other support systems processing sensitive data that are hosted on shared infrastructure (e.g., public cloud) the risk can be greater due to more stakeholders involved and additional ways of accessing the infrastructure, for example, cloud management consoles [18].

For containerized deployments, strict access control at the host level is particularly important. In contrast the virtualized deployments, where each VM runs their own operating system on top of the host machine, containerized workloads only package application software and depend on the host's kernel. For this reason, running containerized workloads with elevated privileges introduces a significant security risk. An attacker may exploit these privileges to perform a container breakout, compromising the host machine and neighboring workloads [19]. While it can be expected that containerized 5G network components would usually not share the same host with external and untrusted tenants, it is good practice to enforce strict guest-host isolation in all cases to limit the potential for lateral movement in the event of a compromise.

5.1.2.2 Integrity (AP2-I)

The threats to data confidentiality due to insufficient access control outlined in AP2-C also apply in this category. Attacks resulting in a breach of confidentiality, such as the compromise of management tools or container and hypervisor breakouts, can just as well result in a breach of integrity.

Another infrastructure component that can result in significant damage to large parts of the network when compromised is the VM or container image registry. Unless properly protected, malicious actors may be able to modify VM or container images of 5G NFs, which may in turn be used to instantiate tampered workloads containing backdoors or other modifications. This could lead to the deployment of malicious software within the network, potentially resulting in further unauthorized access, data breaches, and service disruptions [20]. For example, a compromised User Plane Function instance may lead to a breach of user related data and could allow attackers to expand their attack to other parts of the 5G Core Network.

Moreover, unauthorized physical access to infrastructure components can also greatly increase the risk of tampering with the platform hosting the 5G System. Since these components are likely to expose management interfaces for local operation, this can make it easy for malicious actors to make modifications at the infrastructure level that can cause severe damage to the 5G components hosted on top. (See also [21]).

5.1.2.3 Availability (AP2-A)

The threats to availability due to insufficient access control of infrastructure and MANO components are similar to those outlined in AP2-C and AP2-I. Breaches of logical access control to infrastructure resources, such as container and hypervisor breakout, the compromise of management tools, as well as physical intrusion into data centers may well result in degraded availability for parts of a 5G deployment.

5.1.3 Implementation Flaw (AP3)

Flawed implementations can give rise to a plethora of security vulnerabilities threatening confidentiality, integrity, and availability. While following best practices for secure coding and testing help to reduce the likelihood of implementation issues making it into the final product, they cannot fully prevent such issues from happening, particularly in complex systems such as 5G networks.

5.1.3.1 Confidentiality (AP3-C)

Hard-coded credentials in workload images pose a significant confidentiality risk that is to be avoided during implementation. Virtual Machine (VM) and container images containing unprotected credentials can be easily accessed by malicious users and may allow them unauthorized access not only to workloads instantiated from those images, but potentially to other components connected to them as well. Considering that many new 5G network deployments incorporate cloud environments and shared image repositories accessible from the internet, the risk is even greater and examples for credential breaches are numerous [22] [23].

Flawed workload isolation as described in AP2-C and insecure resource management can also be caused due to implementation failures in hypervisors, container engines, or even hardware components. In the worst case, this can result in a complete VM/container breakout, but other more subtle breaches can also leak sensitive information. For example, Attackers may attempt side-channel attacks by exploiting CPU registers, main memory, or other shared resources to gain unauthorized access to sensitive information [24]. As long as 5G network components are deployed on dedicated hosts and do not share their platform with external, untrusted tenants the risk is limited, but not zero.

Implementation weaknesses in network security protocols may also be exploited by attackers to downgrade, bypass, or completely disable essential security mechanisms in the communication between different Infrastructure components. For example, implementations of the common security protocol TLS that is widely used by infrastructure components have been subject to prominent security weaknesses that could allow attackers to access secret keys and break its encryption [25]. If Management Plane traffic is compromised, it may allow malicious actors to access management commands, credentials, and other sensitive information that could them to further expand their attack.

5.1.3.2 Integrity (AP3-I)

Before a virtualized/containerized 5G NFs is instantiated, its software image should be checked for accidental or malicious tampering by verifying its cryptographic signature. If either this signature validation is flawed or the image has not been signed by the developer in the first place, the integrity of the software cannot be confirmed making it possible for attackers to inject malicious code that could cause severe damage to the 5G deployment [26].

Software vulnerabilities in hypervisors or container engines may allow malicious tenants to break workload isolation and tamper with neighboring workloads or even the host machine. Similarly, kernel vulnerabilities on machines hosting containerized workloads can allow an attacker to perform a container breakout, gaining unauthorized access and control over the host system [27]. Again, this attack path is primarily relevant for components of the 5G System hosted on shared infrastructure.

Vulnerabilities in infrastructure management functions can also lead to serious security breaches. Attackers might exploit unintentional weaknesses or intentional backdoors within these components to compromise virtualized 5G NFs [28]. This is particularly relevant for those components directly interfacing with workloads, for example, monitoring tools.

Similar to AP3-C, implementation weaknesses in network security protocols also threaten the integrity of communications between different Infrastructure components, potentially allowing attackers to tamper with legitimate communication or even injecting malicious traffic [6]. Tampering with Infrastructure management traffic, for example, may lead to incorrect instructions being supplied to NFVI components, incorrect monitoring data, and eventually threaten the stability of the overall network deployment. Furthermore, software images of 5G Network Functions themselves may also be tampered with as they are transferred over the network, posing a similar threat as outlined in AP2-I.

5.1.3.3 Availability (AP3-A)

A threat to availability may result from implementation flaws in the virtualization layer, incl. the hypervisor and container engine, or vulnerabilities affecting the host machine. These issues could either affect the availability of the 5G NF workloads running on top of the virtualization layer directly (e.g., crashing them or otherwise inhibiting their operation) or cause weaknesses in workload isolation, opening the door for malicious workloads to abuse them, for example, by mounting a host-based Denial of Service attack [29] [30].

5.1.4 Specification Weakness (AP4)

Since technical specifications applicable industry-wide —such as those developed by 3GPP— are primarily concerned with ensuring interoperability, their scope is often limited to determining the interactions and communication flows between functional elements, rather than the exact implementation of the elements themselves. As such, this is also where specification weaknesses tend to occur.

5.1.4.1 Confidentiality (AP4-C)

A lack of specification or under-specification of security functionality in virtualization technology can present risks to data confidentiality. Unlike the Technical Specifications for the 3GPP System, which define major functional components, communication flows, and concrete protocols for the entire industry, virtualization technologies are often less strictly specified and largely implementation specific. This may lead to interoperability issues, security control gaps, and can complicate Infrastructure hardening, which may be exploited by attackers to compromise the system.

One essential security functionality that is underspecified and can lead to significant data breaches is secrets management. 5G network deployments already include many Network Functions. If those are implemented in the form of Microservices each having a relatively short lifetime, it can complicate secrets provisioning and storage significantly. Still, popular infrastructure components such as Kubernetes store them as unencrypted base64-encoded strings by default [31].

To the extent that security of the Infrastructure subsystem is standardized, for example, by the ETSI NFV specifications, there is still the potential for specification weaknesses to occur, for example, the use of insecure security protocols and algorithms. A vulnerable protocol may allow malicious actors to downgrade, bypass, or completely disable essential security mechanisms.

5.1.4.2 Integrity (AP4-I)

Similar risks related to specification weaknesses as outlined in AP4-C may also affect the integrity of the Infrastructure subsystem. For example, if a technical specification describes a flawed mechanism for verifying software images before instantiating a virtualized/containerized workload, it may fail to provide the intended integrity protection for the executed code. However, the authors are not aware of concrete examples of such major specification flaws. Rather, it seems more likely that missing security controls due to under-specification and proprietary solutions would lead to integrity issues at the 5G infrastructure.

5.1.4.3 Availability (AP4-A)

Availability failures are not usually a direct result of specification weaknesses. While inadequate or lacking specification certainly does not contribute to the goal of ensuring availability, other phases of the system lifecycle, e.g., design, implementation, and test, are much more important in that respect.

5.1.5 Failure of Power Supply (AP5)

Electric power is essential for any computer system to function. As such, it is up to the system operator to ensure a reliable and consistent supply of power to avoid compromising availability, and to some extent the integrity, of their deployment.

5.1.5.1 Confidentiality (AP5-C)

A breach of confidentiality is usually not a direct damage resulting from a power supply failure. While it is certainly possible that confidentiality is also breached as part of a more complex incident (e.g., during recovering phase involving third-party service providers), the primary damage is expected in the “Availability” category.

5.1.5.2 Integrity (AP5-I)

A power supply failure may result in a compromise of data integrity. For instance, sudden loss of power may affect data-at-rest [32] and could even damage storage devices. However, modern database systems and file systems usually come with appropriate controls to minimize or completely prevent these issues (e.g., transactional processing, checksum confirmation). Hence, the primary damage is expected in the “Availability” category.

5.1.5.3 Availability (AP5-A)

A failure of power supply has the most potential to directly result in a compromise of availability – be it at a limited scale, resulting in a degradation of service quality, or large scale, potentially resulting in a complete unavailability of 5G service.

A *blackout* (i.e., a complete power outage) may be the most obvious scenario of this type of failure. This can be the result of an environmental condition, such as a thunderstorm [33], or intentionally malicious acts against the power infrastructure [34]. If a data center hosting central 5G network components is affected from a blackout and cannot compensate for the loss of power by other means (e.g., batteries, generators), it can have a severe impact on the quality and availability of end-to-end services as well.

A *brownout* describes a condition under which the voltage levels drop to the extent that some electrical equipment can no longer operate. This, too, may be caused incidentally, for example, due to environmental factors, or intentionally by the power provider reducing the load on the electric grid to prevent a blackout. Depending on their severity and the presence of compensating controls by the infrastructure provider, they can still lead to similar detrimental effects to service availability.

Other conditions affecting the electric power supply, such as spikes, surges, and sags usually have a shorter duration. Nevertheless, they can still significantly impact the stability and reliability of the Infrastructure subsystem and thus, the 5G NF workloads running on top of it. In the worst cases, they can result in permanent damage to electronic components, resulting in degraded performance or a complete disruption of service until those parts are replaced [35] [36].

Ideally, 5G network operators should plan their deployments in such a way that isolated incidents do not affect the availability of 5G services, for example, by ensuring the presence of compensating controls (e.g., uninterruptible power supplies, alternative power sources) and distributing 5G NFs geographically over multiple data centers. However, there is still a chance for unforeseen outages to negatively affect network availability, or service quality.

5.1.6 Failure of Hardware Components or Unavailability of Spare Parts (AP6)

Although mobile network operators have gradually moved away from many proprietary technologies in favor of widely available commodity hardware, there is still potential for failures at the physical layer to impact system integrity and availability. It is therefore up to system designers, implementors, and operators, to build the system in such a way that isolated failures can be identified and rectified early on before impacting the 3GPP service layer.

5.1.6.1 Confidentiality (AP6-C)

A breach of confidentiality is usually not a direct damage resulting from a failure of hardware components. While it is certainly possible that confidentiality is also breached as part of a more complex incident (e.g.,

during recovering phase involving third-party service providers), the primary damage is expected in the “Integrity” and “Availability” categories.

5.1.6.2 Integrity (AP6-I)

Unintentional failures

A failure of hardware components may result in a compromise of data integrity. Depending on which system component is affected, it could affect data-in-processing (e.g., due to a CPU or GPU failure), data-in-transit (e.g., due to a damaged network cable), and data-at-rest (e.g., due to bit rot). For example, disk failures can arise from manufacturing defects [37] or environmental factors, such as extreme temperatures, humidity, or physical shocks. In the context of 5G, such issues may result in minor issues such as damaged system configurations, or in the worst case, affect sensitive data-at-rest such as subscriber profiles or billing data.

Issues with network cables, too, can be the result of manufacturing defects, insufficient insulation, or physical damage. Many of these low-level failure scenarios are likely to be mitigated by adequate security controls. For instance, network protocols implement error detection and recovery functionality to compensate for unreliable channels.

Intentional attacks

Besides unintentional hardware failures, there are also attacks aimed at abusing the physical characteristics of specific hardware components. The most prominent example may be *Rowhammer* which targets Dynamic random-access memory modules which almost all computer systems rely on. It has been shown that Rowhammer attacks can be used to compromise data integrity, confidentiality, and overall availability of the affected system [38], even in cloud environments that could be hosting 5G network components.

5.1.6.3 Availability (AP6-A)

Failures of the essential hardware components of the Infrastructure subsystem —i.e., physical compute, storage, and network equipment— can all have a negative effect on service availability. Strategies to prevent a single hardware failure from impacting the 3GPP service layer, for example, the 5G network functions running on top, include physical redundancy, error detection, and backup and recovery mechanisms. Ideally, an isolated incident that does not affect multiple physical components at a large scale can be mitigated without severe impact to higher layers of the system.

Besides issues with the compute, storage, and networking components, failure of supporting infrastructure, such as data center cooling systems, can also threaten the availability of the 5G network infrastructure. A potential overheating of electronic components, for instance, can lead to severe denial of service situations [35]. Such failures may result from intentional sabotage or unintentional incidents, incl. technical malfunctions or human errors.

5.1.7 System Design & Architecture Flaws (AP7)

Although guided by technical specifications, the system design and architecture of 5G networks is still defined by the network operator to a large extent. At this stage of the system lifecycle, too, there are pitfalls that can impact the security of the overall deployment.

5.1.7.1 Confidentiality (AP7-C)

Insufficient network segregation can significantly increase the risk of security breaches and unintentional malfunctions spreading throughout the 5G deployment. For example, without internal segregation between environments of different levels of sensitivity, attackers may be able to expand an initial compromise by moving laterally within the network and eventually reaching highly sensitive workloads (e.g., subscriber databases, business support systems). This is particularly relevant to distributed 5G deployments which may employ a mix of centralized data centers, smaller Edge Computing locations (which may feature weaker physical security), and cloud environments (which may be shared with other tenants). As these deployments do not always have a clearly defined network perimeter, enforcing defense in depth is even more crucial.

5.1.7.2 Integrity (AP7-I)

The threats to integrity due to design and architecture related flaws are like those outlined in AP7-C. Lateral movement by attackers may well result in a breach of data integrity in the 5G System.

5.1.7.3 Availability (AP7-A)

Failure to avoid single points of failure and to ensure defence in depth can expose a 5G deployment and its operator to significant risks. As already outlined in AP5-A and AP6-A, a lack of redundancy in critical areas such as power infrastructure and physical network infrastructure, makes the overall deployment vulnerable to outages due to unintentionally defects or attacks. The same also applies to higher system layers where reliance on a single technology could lead to widespread service disruptions. For example, if the entire 5G network relies on one particular virtualization technology or critically depends on the availability of one particular cloud service provider, the availability of 5G service may be at risk of either of them experiences a critical vulnerability or similar issues.

Additionally, insufficient traffic isolation poses a risk to the ability manage the 5G infrastructure. Transferring management traffic over the same channels as 5G User Plane and Control Plane traffic can lead to the degradation of management communications and, in the worst case, a complete loss of control when management interfaces cannot be reached.

5.2 Public Mobile Network Operator (MNO) Perspective

This section discusses the risk scenarios related to the MNOs perspective.

For each subsection following, a short evaluation of the relevance of the risk scenarios in the Infrastructure subsystem is discussed and the probability that potential attack paths could be exploited is derived.

5.2.1 Emergency Call (TKG § 164)

Security of the Infrastructure subsystem affects all services provided by the 5G System. However, there is no risk that would threaten Emergency Calls in particular. Instead, this is just one of the many services whose availability would be affected by a compromise of the infrastructure (see also section 5.2.6).

Therefore, **the Infrastructure subsystem is not considered** as a relevant attack domain for the risk scenario Emergency Call services.

5.2.2 Cell Broadcast (TKG § 164a)

Security of the Infrastructure subsystem affects all services provided by the 5G System. However, there is no risk that would threaten Cell Broadcast Services in particular. Instead, this is just one of the many services whose availability would be affected by a compromise of the infrastructure (see also section 5.2.6).

Therefore, **the Infrastructure subsystem is not considered** as a relevant attack domain for the risk scenario Cell Broadcast Services.

5.2.3 Telecommunications Secrecy (TTDSG § 3)

Telecommunication Secrecy requires user data to be kept confidential. Since the Infrastructure subsystem comprises the physical networking equipment and storage media that transport and retain user data of the 5G System, security incidents within this subsystem have the potential to threaten the confidentiality of this information and expose it to unauthorized third parties.

Therefore, **the Infrastructure subsystem is considered** as a relevant attack domain for the risk scenario Telecommunication Secrecy. The attack paths are similar to those that may lead to Subscriber Data Leaks described in section 5.2.9.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	Misconfigurations in the Infrastructure subsystem threatening user related data primarily affects data-at-rest. If the public MNO or its third-party infrastructure providers fail to setup/utilize fundamental security controls, such as storage encryption and secure key management, attackers may be able to gain unauthorized access to user data. Security hardening of internal NFVI components and external cloud environments is another important configurational measure. Failure to minimize their attack surface can result in exposure of internal interfaces and services that may allow attackers to gain unauthorized access to sensitive data. The probability of these incidents is considered HIGH . Although public network operators should have sufficiently mature testing and change management processes in place to prevent such breaches of confidentiality, there are a number of real-world examples of when such measures have fallen short of providing adequate protection [39] [40].	High
Insufficient Access Controls	Confidentiality (AP2-C)	Access control to the Infrastructure subsystem comprises logical access (e.g., to management or NFVI components) and physical access (i.e. to data centers). A gap in either of these may lead to a breach in telecommunication secrecy. Insufficiently restricted logical access to internal systems of public MNOs or those of their service providers may allow attackers to access communication contents or metadata. For example, accounts at a cloud service provider requiring only single-factor authentication or no authentication at all may be easily bypassed by malicious actors [18]. Assignment of overly broad logical access rights to internal and third-party staff can further increase the likelihood of an incident. There have been reports about breaches involving employees abusing their access for personal gain, which may have been prevented with stricter access control and monitoring [41] [42]. Similar attacks would also be possible with access to the Infrastructure subsystem. The probability of these incidents is considered HIGH , as large-scale public network deployments involve a lot of stakeholders for management and maintenance activities – including third parties if parts of the infrastructure are provided by external data centers or cloud service providers – for all of which the principle of least privilege should to be enforced.	High

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	Implementation flaws threatening user related data may also occur in the Infrastructure subsystem. For example, vulnerabilities in the virtualization layer may allow attackers to breach the horizontal isolation between workloads, or even the vertical isolation between workloads and host [43]. If an attacker is able to gain access to components processing communication contents this way, they could also breach telecommunication secrecy. Beyond the scope of a single host, implementation flaws allowing attackers to gain unauthorized access to management components can pose an even greater threat. If attackers are able to compromise MANO functions, they would likely be able to breach confidentiality –and other protection goals– of several workloads [44]. The probability of these incidents is considered MEDIUM . Generally, it is expected that central network components handling user’s communication contents and metadata (e.g., UDM/HSS, UPF) would be implemented in/hosted on dedicated hardware rather than on infrastructure shared with other tenants, making it more difficult for attackers to breach confidentiality at the infrastructure layer.	Medium
Specification Weakness	Confidentiality (AP4-C)	Although the technical specifications focus primarily on network interface security and associated procedures, there are certain aspects concerning the protection of data-at-rest and thus, the Infrastructure subsystem. However, considering the specification and review process that involves numerous industry stakeholders and, at times, even public review, an error occurring at this stage is rather unlikely. On the other hand, data-in-transit is usually protected at higher system layers by 5G Network Functions hosted on top of the infrastructure, e.g., by means of TLS or IPSec. Therefore, the probability of these incidents is considered LOW .	Low
Failure of Power Supply	Confidentiality (AP5-C)	A breach of confidentiality is usually not a direct damage resulting from a power supply failure. While it is certainly possible that confidentiality is also breached as part of a more complex incident (e.g., during recovering phase involving third-party service providers), the primary damage is expected in the “Availability” category. Therefore, the probability of these incidents is considered LOW .	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	A breach of confidentiality is usually not a direct damage resulting from a hardware failure. While it is certainly possible that confidentiality is also breached as part of a more complex incident (e.g., during recovering phase involving third-party service providers), the primary damage is expected in the “Availability” category. Therefore, the probability of these incidents is considered LOW .	Low
Design and architecture flaws	Confidentiality (AP7-C)	A failure of Telecommunication secrecy may also be the result of a network design/architecture failure leading to insufficient confidentiality protection of user related data. For example, if important but optional security controls are omitted (e.g., physical security gateways providing protection at the network layer) attackers may be able to access communication contents. However, it is expected that such blatant design flaws would be prevented by a public MNO’s security review processes. Therefore, the probability of these incidents is considered LOW .	Low

5.2.4 Emergency Preparedness (TKG § 184)

Security of the Infrastructure subsystem affects all services provided by the 5G System. However, there is no risk that would threaten Emergency Preparedness in particular. Instead, this is just one of the many services whose availability would be affected by a compromise of the infrastructure (see also section 5.2.6).

Therefore, **the Infrastructure subsystem is not considered** as a relevant attack domain for the risk scenario Emergency Preparedness.

5.2.5 Regional Network Failure

Regional Network Failure describes network unavailability limited to a relatively small geography. This is a less severe form of the Multi-Regional Network Failure described in section 5.2.6. Since the related attack paths are similar and incidents in central parts of the infrastructure would almost certainly affect a larger service area (in contrast to failure of a singular base station, for example), the reader is referred to the next section.

Therefore, **the Infrastructure subsystem is not considered** as a relevant attack domain for the risk scenario Regional Network Failure.

5.2.6 Multi-Regional Network Failure

Multi-Regional Network Failure describes network unavailability in large parts of the service area. Considering that the Infrastructure subsystem outlined in section 4 supports central network components, such as 5G Core Network Functions, IMS, and other elements, there is a considerable risk that an incident affecting this subsystem would impact large areas of the network.

Therefore, **the Infrastructure subsystem is considered** as a relevant attack domain for the risk scenario Multi-Regional Network Failure.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	Misconfiguration can happen at multiple points of the Infrastructure subsystem (e.g., MANO components and interfaces, NFVI components and interfaces) and lead to a degradation of service quality or a complete availability failure. Situations where misconfigurations can occur are numerous, incl. regular maintenance [45] and rollout of new hardware [46] which have led to outages in the past, even with major cloud service providers. Although these organization are expected to have sufficiently mature maintenance and change processes in place to prevent such incidents, this does not eliminate the risk entirely. Therefore, the probability of these incidents is considered MEDIUM .	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Availability (AP2-A)	Access control to the Infrastructure subsystem comprises logical access (e.g., to management or NFVI components) and physical access (i.e. to data centers). A gap in either of these may lead to a large-scale failure of availability. Assignment of overly broad logical access rights to internal staff can increase the likelihood of an incident, e.g., when a network administrator also has access to the platform on which Network Functions are hosted. This includes the timely revocation of access rights to prevent current or former employees from intentional acts of sabotage [47]. The probability of these incidents is considered MEDIUM , as large-scale public network deployments involve a lot of stakeholders for management and maintenance activities—including third parties if parts of the infrastructure are provided by external data centers or cloud service providers—for all of which the principle of least privilege should be enforced.	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation failures affecting NFVI and MANO components in a central data center may result in a degradation of service quality or complete availability failures in large parts of the network. For example, unless virtualization layer effectively manages resource quotas of the workloads deployed on top of the infrastructure and enforces limits where necessary, individual workloads may consume more resources than expected, affecting neighbouring workloads. In the most severe case, implementation failures may even cause a complete crash or Denial of Service situation that affects all hosted workloads. The probability of these incidents is considered MEDIUM , as severe issues should be caught by testing processes of both the vendor and the MNO in most cases. Even though not all implementation flaws may result in a complete outage [48], they still have the potential to negatively impact the performance of services in a large area of the network. In addition, it is expected that central network components would be implemented in/hosted on dedicated hardware rather than on infrastructure shared with other tenants, reducing the likelihood of neighbouring tenants intentionally harming availability at the infrastructure layer.	Medium
Specification Weakness	Availability (AP4-A)	Availability failures are generally not a direct result of a specification weakness. For ensuring availability, design, implementation, and tests are much more relevant (see “Design and architecture flaws” below). Therefore, the probability of these incidents is considered LOW .	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Availability (AP5-A)	Failure of power supply in a central data center hosting key network components can impact the availability of large parts of the network. For example, unavailability of any given 5G Core Network Function would likely affect subscribers in major parts of a public MNO's service area. The reason power outages may include unintentional defects [49], intentional sabotage, external issues at the power grid operator [50], or environmental conditions [33]. Although public MNOs would generally have adequate controls in place to prevent these large-scale incidents from happening (e.g. redundant access to power grids, uninterruptable power supplies, generators), not all aspects of the power supply are under the control of the MNO itself, and the numerous real-world examples of power failures leading to data center outages exemplifies that the risk is significant. Therefore, the probability of these incidents is considered MEDIUM .	Medium
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	Failure of hardware components at a central point of the network can impact the availability of large parts of the 5G deployment. Similar to a failure of power supply, public MNOs would generally have redundancy and recovery capacity in place to mitigate the most severe incidents from happening within their own facilities. However, this scenario considers a failure of any physical components –incl. compute, storage, and network infrastructure– which are not just deployed within the data center. For example, a backbone fiber cable outside of the data center may be unintentionally interrupted, negatively affecting service quality and/or availability [51] [52]. Therefore, the probability of these incidents is considered MEDIUM .	Medium
Design and architecture flaws	Availability (AP7-A)	Various design decisions in the Infrastructure subsystem can affect the availability/resiliency of the overall network. This starts with the geographical location and distribution of the data centers hosting the compute and network infrastructure. Ideally, a regional incident, such as a natural catastrophe or an outage should not impact multiple locations at once, and public MNOs should have the ability to compensate for a local availability issue by adequate recovery sites and the ability to switch over. However, large-scale events may still impact service availability and quality [53]. Inside the data center, facility and system design also plays a crucial role in ensuring availability. Redundancy in power supply, communication infrastructure, and physical compute/storage infrastructure should be a key consideration when planning a public network deployment. Failure to consider these points during network design and deployment has a MEDIUM likelihood of resulting in a Multi-Regional Network Failure.	Medium

5.2.7 Failure of Mobile Service

Failure of Mobile Service describes a scenario in which specific communication services are unavailable in the entire coverage area, even though the user equipment is successfully registered and connected to the network. Considering that any severe issue at the infrastructure layer would likely impact all services provided by the 3GPP System, such a scenario is unlikely to be caused by this particular subsystem. Instead, a major problem would more likely result in a complete unavailability of entire network areas, as described in section 5.2.6.

Therefore, **the Infrastructure subsystem is not considered** as a relevant attack domain for the risk scenario Failure of Mobile Service.

5.2.8 Fraud

Fraud describes acts of abuse for monetary or other gains. In the context of telecommunications, this often results in fraudulent service usage or fraudulent charges to operators or their subscriber's falling victim to fraud. This, however, concerns primarily the 3GPP service layer, incl. 5G Core and IMS, but not the Infrastructure subsystem.

Therefore, **the Infrastructure subsystem is not considered** as a relevant attack domain for the risk scenario Fraud.

5.2.9 Subscriber Data Leak

Subscriber data leaks happen if personal identifiable data of users is exposed to unauthorized third parties. Considering that the Infrastructure subsystem comprises the physical networking equipment and storage media that transport and retains subscriber related data processed by the 5G System, there is a risk for attacks resulting in such exposure.

Therefore, **the Infrastructure subsystem is considered** as a relevant attack scenario for the risk scenario Subscriber Data Leaks.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	Misconfigurations in the Infrastructure subsystem threatening user related data primarily affects data-at-rest. If the public MNO or its third-party infrastructure providers fail to setup/utilize fundamental security controls, such as storage encryption and secure key management, attackers may be able to gain unauthorized access to subscriber data. Security hardening of infrastructure components, incl. external cloud environments, is another important configurational measure. Failure to minimize the attack surface of NFVI resources (e.g., storage systems, network infrastructure) and MANO components, such as Element Managers, can result in exposure of internal interfaces and services that may allow attackers to gain unauthorized access to sensitive data [39]. The probability of these incidents is considered HIGH . Although public network operators should have sufficiently mature testing and change management processes in place to prevent such breaches of confidentiality, there are a number of real-world examples of when such measures have fallen short of providing adequate protection.	High
Insufficient Access Controls	Confidentiality (AP2-C)	Access control to the Infrastructure subsystem comprises logical access (e.g., to management or NFVI components) and physical access (i.e. to data centers). A gap in either of these may lead to a subscriber data leak. Insufficiently restricted logical access to internal systems of public MNOs or those of their service providers may allow attackers to access communication contents or metadata. For example, accounts at a cloud service provider requiring only single-factor authentication or no authentication at all may be easily bypassed by malicious actors [18] [40]. Assignment of overly broad logical access rights to internal and third-party staff can further increase the likelihood of an incident. There have been reports about breaches involving employees abusing their access for personal gain, which may have been prevented with stricter access control and monitoring [41] [42]. Similar attacks would also be possible with access to the Infrastructure subsystem. The probability of these incidents is considered HIGH , as large-scale public network deployments involve a lot of stakeholders for management and maintenance activities—including third parties if parts of the infrastructure are provided by external data center or cloud service providers—for all of which the principle of least privilege should to be enforced.	High

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	Implementation flaws threatening subscriber related data may also occur in the Infrastructure subsystem. For example, vulnerabilities in the virtualization layer may allow attackers to breach the horizontal isolation between workloads, or even the vertical isolation between workloads and host [43]. If an attacker is able to gain access to components processing user related data, it could result in a subscriber data leak. Beyond the scope of a single host, implementation flaws allowing attackers to gain unauthorized access to management components can pose an even greater threat. If attackers are able to compromise MANO functions, they would likely be able to breach confidentiality –and other protection goals– of several workloads [44]. The probability of these incidents is considered MEDIUM . Generally, it is expected that central network components handling user’s communication contents and metadata (e.g., UDM/HSS, UPF) would be implemented in/hosted on dedicated hardware rather than on infrastructure shared with other tenants, making it more difficult for attackers to breach confidentiality at the infrastructure layer.	Medium
Specification Weakness	Confidentiality (AP4-C)	Although the technical specifications focus primarily on network interface security and associated procedures, there are certain aspects concerning the protection of data-at-rest that may threaten the secrecy of subscriber data in the Infrastructure subsystem. For example, TS 33.501 specifies certain requirements on the protection from physical attacks and secure environments in certain 5G NFs, such as the UDM which stores subscriber profiles (see [54], section 5.8.1). However, considering the specification and review process that involves numerous industry stakeholders and, at times, even public review, an error occurring at this stage is rather unlikely. On the other hand, when subscriber data is transferred over the network, it is usually protected at higher system layers by 5G Network Functions hosted on top of the infrastructure, e.g. by means of TLS or IPsec. Therefore, the probability of these incidents is considered LOW .	Low
Failure of Power Supply	Confidentiality (AP5-C)	A breach of confidentiality is usually not a direct damage resulting from a power supply failure. While it is certainly possible that confidentiality is also breached as part of a more complex incident (e.g., during recovering phase involving third-party service providers), the primary damage is expected in the “Availability” category. Therefore, the probability of these incidents is considered LOW .	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	A breach of confidentiality is usually not a direct damage resulting from a hardware failure. While it is certainly possible that confidentiality is also breached as part of a more complex incident (e.g., during recovering phase involving third-party service providers), the primary damage is expected in the “Availability” category. Therefore, the probability of these incidents is considered LOW .	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Design and architecture flaws	Confidentiality (AP7-C)	A breach of subscriber related data may also be the result of a network design/architecture failure leading to insufficient confidentiality protection of user data. For example, if important but optional security controls are omitted (e.g., physical security gateways providing protection at the network layer) attackers may be able to access subscriber data. However, it is expected that such blatant design flaws would be prevented by a public MNO's security review processes. Therefore, the probability of these incidents is considered LOW .	Low

5.3 User Perspective

This section describes different risk scenarios from the perspective of a user of the public mobile network. Private 5G networks are not in scope of this section. Since some of the risk scenarios one might typically expect here (e.g. emergency calls) are already described in the previous paragraphs, they won't be mentioned again in this section.

5.3.1 Urgent Telephone Calls Not Possible

Although unavailability of the Infrastructure subsystem can negatively affect the 5G and IMS functions required for enabling telephone calls, the potential impact would likely not be limited to one service only, such as voice services. Related attack paths are described in section 5.2.6.

Therefore, **the Infrastructure subsystem is not considered** as a relevant attack domain for the risk scenario Urgent Telephone Calls Not Possible.

5.3.2 Authenticity of the Receiver of a Call

Although the Infrastructure subsystem supports the relevant 5G and IMS functions enabling voice call services, it is not directly involved in ensuring the authenticity of called party.

Therefore, **the Infrastructure subsystem is not considered** as a relevant attack domain for the risk scenario Authenticity of the Receiver of a Call.

5.3.3 Integrity of Transmitted Data

User data transported or otherwise processed by the 5G System may also be tampered with at lower system layers associated to the Infrastructure subsystem. Even though the 3GPP System specifies relevant security controls, they usually do not cover all areas of an operator's environment processing user data, or they may not even be completely enforced in real-world deployments.

Therefore, **the Infrastructure subsystem is considered** as a relevant attack domain for the risk scenario Integrity of Transmitted Data.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Integrity (AP1-I)	This risk scenario focuses on authenticity/integrity of data by the subscriber when transmitted over the 5G System, i.e., User Plane data. User Plane data-in-transit is specified to be protected at the 3GPP layer. That is, protection should be provided not by the Infrastructure subsystem but by the 5G Network Functions deployed on top. As such, if higher system layers are configured properly, a misconfiguration occurring at the Infrastructure should not impair the authenticity/integrity of transmitted User Plane data (even if an attack is attempted, higher layer controls should effectively mitigate it). However, misconfigurations may happen at those higher layers as well, and it is always recommended to ensure multi-layer security (e.g., by additionally providing integrity protection at the infrastructure layer, for example, by means of IPsec). Therefore, the probability of these incidents is considered MEDIUM .	Medium
Insufficient Access Controls	Integrity (AP2-I)	This risk scenario focuses on authenticity/integrity of data by the subscriber when transmitted over the 5G System, i.e., User Plane data. User Plane data-in-transit is specified to be protected at the 3GPP layer. That is, protection is provided not by the Infrastructure subsystem but by the 5G Network Functions deployed on top. Insufficient access control to infrastructure components hosting 5G NFs that terminate security protocols, such as IPsec and TLS (e.g., UPF), may allow malicious actors to tamper with User Plane data as it is processed by said NFs. However, given that the relevant 5G NFs (e.g., UPF) are generally deployed in a central location in the network protected by multiple layers of logical access control, and the surrounding data center is expected to feature physical security measures, the probability of such an attack is limited. Therefore, the probability of these incidents is considered MEDIUM .	Medium
Implementation Flaw (Quality of the Product)	Integrity (AP3-I)	This risk scenario focuses on authenticity/integrity of data by the subscriber when transmitted over the 5G System, i.e., User Plane data. User Plane data-in-transit is specified to be protected at the 3GPP layer. That is, protection is provided not by the Infrastructure subsystem but by the 5G Network Functions deployed on top. As such, an implementation flaw affecting Infrastructure components should not impair the authenticity/integrity of transmitted User Plane data (even if certain parts do show unexpected behaviour, higher layer controls should effectively mitigate it). However, those higher layers may not always offer the expected protection, and according to the security principle of multi-layer security, it is recommended to also provide integrity protection at the infrastructure layer. In addition, implementation flaws affecting hardware security components, such as Hardware Security Modules (HSM) may expose sensitive key material to attackers, which may in turn be used to break cryptographic security controls. While such implementation flaws are certainly possible, these infrastructure-layer attacks may be quite difficult to realize, particularly if the malicious actors are not running software on the same host as the 5G component under attack, or do not have physical access. The probability of these incidents is considered LOW .	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Specification Weakness	Integrity (AP4-I)	This risk scenario focuses on authenticity/integrity of data by the subscriber when transmitted over the 5G System, i.e. User Plane data. User Plane data-in-transit is specified to be protected at the 3GPP layer. That is, protection is provided not by the Infrastructure subsystem but by the 5G Network Functions deployed on top. As such, weaknesses in the technical specifications affecting the integrity of the 5G infrastructure itself may also lead to authenticity/integrity issues at the 3GPP service layer. For example, if the technical standards would not specify the integrity protection of VM / container images, implementors may omit this security control, increasing the likelihood of tampered NF instances being deployed, eventually affecting authenticity/integrity of User data. However, the probability of such an attack is considered LOW , and such blatant control gaps are expected to be identified during either design, implementation, or test of the system.	Low
Failure of Power Supply	Integrity (AP5-I)	An integrity failure of data-in-transit is usually not a direct damage resulting from a power supply failure. While it is certainly possible that data-in-transit may be lost due to an unexpected power issue, the primary damage is expected in the “Availability” category. Therefore, the probability of these incidents is considered LOW .	Low
Failure of Hardware Components or Unavailability of Spare Parts	Integrity (AP6-I)	An integrity failure of data-in-transit is usually not a direct damage resulting from a hardware failure. While it is certainly possible that data-in-transit may be lost or otherwise damaged due to an unexpected hardware issue, the primary damage is expected in the “Availability” category. Even if there are certain issues at the infrastructure layer that may affect the transported data (e.g., a faulty cable), higher controls should mitigate any impact on the 3GPP service layer. Therefore, the probability of these incidents is considered LOW .	Low
Design and architecture flaws	Integrity (AP7-I)	No specific attack paths have been identified.	Low

5.3.4 User’s Identity

The Infrastructure subsystem hosting 5G Network Functions and other mobile network operator systems is directly involved in processing and storing various user identities, such as SUPI, 5G GUTI, and other personal identifiable information. Therefore, there is also a risk of this information being breached and **the Infrastructure subsystem is considered** as a relevant attack domain for the risk scenario User’s Identity.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Integrity (AP1-I)	Misconfigured Infrastructure components have the potential to leak user identities. For example, cloud storage that is not properly secured may be accessed by attackers and leak such sensitive information. This is not just applicable to 5G Network Functions themselves, but also to other support systems (e.g., BSS) that store or otherwise process user identities [39]. In addition, Infrastructure management components (e.g., VNF Managers, Element Managers) that are either misconfigured or insufficiently hardened and that have access to the 5G Network (e.g., for monitoring purpose) may also allow unauthorized parties to access user identities. Therefore, the probability of these incidents is considered HIGH .	High
Insufficient Access Controls	Integrity (AP2-I)	Insufficient access control to (virtual) network infrastructure and associated management components may allow internal or external attackers to gain unauthorized access to user identities. Similar to “Misconfigurations”, insufficient access control may affect not just the 5G NFs themselves, but also management components, cloud storage buckets, and other connected applications. Considering the potentially large number of stakeholders involved in the provisioning of 5G services, particularly in virtualized deployments (incl. platform admin, 5G NF admin, MANO operator, etc.) as well as the number of reported incidents related to cloud deployments, the probability of these incidents is considered HIGH . (See also [18]).	High
Implementation Flaw (Quality of the Product)	Integrity (AP3-I)	Implementation flaws in Infrastructure components may allow attackers to bypass or break security controls and gain unauthorized access to user identities. Such implementation issues may include flaws in workload isolation at the virtualization layer allowing attacks between neighbouring workloads or isolation flaws at the HW layer (e.g., Meltdown/Spectre). While such implementation flaws are certainly possible, these infrastructure-layer attacks may be quite difficult to realize, particularly if the malicious actors are not running software on the same host as the 5G component that is under attack. Therefore, the probability of these incidents is considered LOW .	Low
Specification Weakness	Integrity (AP4-I)	Although the technical specifications focus primarily on network interface security and associated procedures, there are certain aspects concerning the protection of data-at-rest and thus, the Infrastructure subsystem. However, considering the specification and review process that involves numerous industry stakeholders and, at times, even public review, an error occurring at this stage is rather unlikely. On the other hand, data-in-transit is usually protected at higher system layers by 5G Network Functions hosted on top of the infrastructure, e.g., by means of TLS or IPSec. Therefore, the probability of these incidents is considered LOW .	Low
Failure of Power Supply	Integrity (AP5-I)	A breach of confidentiality is usually not a direct damage resulting from a power supply failure. While it is certainly possible that confidentiality is also breached as part of a more complex incident (e.g., during recovering phase involving third-party service providers), the primary damage is expected in the “Availability” category. Therefore, the probability of these incidents is considered LOW .	Low
Failure of Hardware Components or Unavailability of Spare Parts	Integrity (AP6-I)	A breach of confidentiality is usually not a direct damage resulting from a hardware failure. While it is certainly possible that confidentiality is also breached as part of a more complex incident (e.g., during recovering phase involving third-party service providers), the primary damage is expected in the “Availability” category. Therefore, the probability of these incidents is considered LOW .	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Design and architecture flaws	Integrity (AP7-I)	No specific attack paths have been identified.	Low

5.3.5 Eavesdropping of a Communication

As outlined in the previous section, the Infrastructure subsystem by necessity is involved in processing user data, incl. communication contents. As such, there is also a risk of this data being breached by malicious actors at lower system layers.

So, the **Infrastructure subsystem is considered** as a relevant attack domain for the risk scenario Eavesdropping of a Communication.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	This risk scenario focuses on the confidentiality of User Plane data-in-transit. That contrasts with the risk scenario “Telecommunication Secrecy”, which also covers metadata on user’s communication which may be retained in the 5G System and associated support systems. User Plane data-in-transit is specified to be protected at the 3GPP layer. That is, protection is provided not by the Infrastructure subsystem but by the 5G Network Functions deployed on top. As such, a misconfiguration occurring at the Infrastructure should not impair the confidentiality of transmitted User Plane data (even if an attack is attempted, higher layer controls should effectively mitigate it). However, misconfigurations may happen at those higher layers as well, and it is always recommended to ensure multi-layer security (e.g., by additionally providing confidentiality protection at the infrastructure layer, for example, by means of IPSec). Therefore, the probability of these incidents is considered MEDIUM .	Medium
Insufficient Access Controls	Confidentiality (AP2-C)	This risk scenario focuses on the confidentiality of User Plane data-in-transit. User Plane data-in-transit is specified to be protected at the 3GPP layer. That is, protection is provided not by the Infrastructure subsystem but by the 5G Network Functions deployed on top. Insufficient access control to infrastructure components hosting 5G NFs that terminate security protocols (e.g., IPsec, TLS) may allow malicious actors to access User Plane data as it is processed by said NFs. Assignment of overly broad logical access rights to internal and third-party staff can further increase the likelihood of an incident. There have been reports about breaches involving employees abusing their access for personal gain, which may have been prevented with stricter access control and monitoring [41] [42]. Similar attacks would also be possible with access to the Infrastructure subsystem. However, given that the relevant 5G NFs (e.g., UPF) are generally deployed in a central location in the network and the surrounding data center is expected to feature physical security measures limiting probability of such an attack. Therefore, the probability of these incidents is considered MEDIUM .	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	User Plane data-in-transit is specified to be protected at the 3GPP layer. That is, protection is provided not by the Infrastructure subsystem but by the 5G Network Functions deployed on top. As such, an implementation flaw affecting Infrastructure components should not impair the confidentiality of transmitted User Plane data. However, those higher layers may not always offer the expected protection, and according to the security principle of multi-layer security, it is recommended to also provide confidentiality protection at the infrastructure layer. In addition, implementation flaws affecting hardware security components, such as HSMs may expose sensitive key material to attackers, which may in turn be used to break cryptographic security controls. While such implementation flaws are certainly possible, these infrastructure-layer attacks may be quite difficult to realize, particularly if the malicious actors are not running software on the same host as the 5G component under attack, or do not have physical access. Therefore, the probability of these incidents is considered LOW.	Low
Specification Weakness	Confidentiality (AP4-C)	Although the technical specifications focus primarily on network interface security and associated procedures, there are certain aspects concerning the protection of data-at-rest and thus, the Infrastructure subsystem. However, considering the specification and review process that involves numerous industry stakeholders and, at times, even public review, an error occurring at this stage is rather unlikely. On the other hand, data-in-transit is usually protected at higher system layers by 5G Network Functions hosted on top of the infrastructure, e.g., by means of TLS or IPsec. Therefore, the probability of these incidents is considered LOW.	Low
Failure of Power Supply	Confidentiality (AP5-C)	A breach of confidentiality is usually not a direct damage resulting from a power supply failure. While it is certainly possible that confidentiality is also breached as part of a more complex incident (e.g., during recovering phase involving third-party service providers), the primary damage is expected in the “Availability” category. Therefore, the probability of these incidents is considered LOW.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	A breach of confidentiality is usually not a direct damage resulting from a hardware failure. While it is certainly possible that confidentiality is also breached as part of a more complex incident (e.g., during recovering phase involving third-party service providers), the primary damage is expected in the “Availability” category. Therefore, the probability of these incidents is considered LOW.	Low
Design and architecture flaws	Confidentiality (AP7-C)	Eavesdropping of a Communication may also be the result of a network design/architecture failure leading to insufficient confidentiality protection of user related data. For example, if important but optional security controls are omitted (e.g., physical security gateways providing protection at the network layer) attackers may be able to access communication contents. However, it is expected that such blatant design flaws would be prevented by a MNO's security review processes. Therefore, the probability of these incidents is considered LOW.	Low

5.3.6 Slowing Down/Restricting the Connection

An incident affecting the availability of the Infrastructure subsystem would likely have a more severe impact than slowing down or restricting an individual connection. (See also section 5.2.6).

Therefore, **the Infrastructure subsystem is not considered** as a relevant attack domain for the risk scenario Slowing Down/Restricting the Connection.

5.3.7 Connection Failure

An incident affecting the availability of the infrastructure subsystem would likely have a more severe impact than the failure of an individual connection. (See also section 5.2.6).

Therefore, **the Infrastructure subsystem is not considered** as a relevant attack domain for the risk scenario Connection Failure.

5.3.8 Tapping or Altering of the Geographical Position

Although the Infrastructure subsystem hosts 5G components that may store or otherwise process the geographical position of subscribers, it is unlikely that a compromise would affect specifically this data. Relevant attack paths are similar to those in the more generic scenario of a breach of Telecommunications Secrecy, described in section 5.2.3.

Therefore, **the Infrastructure subsystem is not considered** as a relevant attack domain for the risk scenario Tapping or Altering of the Geographical Position.

5.4 Private 5G Network Operator Perspective

Note: The following assumptions apply for these risk scenarios for the private 5G network perspective.

- This scenario is only focussed on on-premise networks.
- The network's backbone is part of the organization's general IT network.
- The network is separated by a firewall, which is part of the 5G network, from the public network.
- The network does not have Multi Access Edge Computing (MEC) or an IP Multimedia Subsystem (IMS).
- This perspective considers networks that are limited to dedicated locations.
- Organizations might operate multiple locations with separate private 5G networks.
- The members of the organization operating the network are also the user of the network. There is no separate group of users or a separate operator.

5.4.1 Network Failure

The infrastructure subsystem provides the essential compute, storage, and networking resources to the workloads deployed on top of it. As such, the availability of the 5G System critically depends on the availability of this subsystem.

Therefore, the **Infrastructure subsystem is considered** as a relevant attack scenario for the risk scenario Network Failure.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	Misconfigurations of the Infrastructure subsystem can have a negative effect on the 3GPP service layer. The likelihood of a private 5G operator misconfiguring the 5G Infrastructure leading to service availability issues may be higher than for a public MNO due to limited resources available for 5G infrastructure configuration management (incl. creating, testing, auditing configuration settings). Additionally, a lack of clear segregation between 5G network infrastructure and generic IT infrastructure can lead to misconfigurations in one domain affecting the other. However, the probability of such severe misconfigurations that would result in a complete network failure is considered MEDIUM .	Medium
Insufficient Access Controls	Availability (AP2-A)	Access control to the Infrastructure subsystem comprises logical access (e.g., to management or NFVI components) and physical access (i.e. to data centers). A gap in either of these may lead to a large-scale failure of availability. Assignment of overly broad logical access rights to internal staff can increase the likelihood of an incident, e.g., when a network administrator also has access to the platform on which Network Functions are hosted. This includes the timely revocation of access rights to prevent current or former employees from intentional acts of sabotage [47]. The probability of these incidents is considered MEDIUM , since especially smaller organizations operating a private 5G network may not always have sufficient resources to enforce strict segregation of roles, necessitating the assignment of broader administrative rights.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws in the Infrastructure subsystem (e.g., Hardware, Operating System, Hypervisor) may lead to negative effects at the 3GPP service layer if they impact the performance of the software or virtualized workloads running on top of it. Ideally, severe issues should be caught by the testing processes of the technology provider or the private network operator. Even if they go unnoticed, very few implementation flaws are expected to result in a complete unavailability of the Infrastructure subsystem and thus, a network outage [48]. Therefore, the probability of these incidents is considered LOW .	Low
Specification Weakness	Availability (AP4-A)	Availability failures are generally not a direct result of a specification weakness. For ensuring availability, design, implementation, and test are much more relevant (see “Design and architecture flaws” below). Therefore, the probability of these incidents is considered LOW .	Low
Failure of Power Supply	Availability (AP5-A)	Smaller, private 5G deployments may not have the resources or expertise to maintain a highly redundant network deployment, distributed across multiple data centers or regions. As such, a power supply failure at a company data center or that of a third-party cloud service provider hosting central network components (e.g., 5G Core) may have a significant impact on the availability of the overall network. Although mature data centers should have controls in place to prevent outages due to a temporary failure of power supply (e.g., UPS, generators), such incidents can still happen. Therefore, the probability of these incidents is considered MEDIUM .	Medium
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	Similar to the case of a “Failure of Power Supply”, private deployments are expected to feature less redundancy and failover capacity that would limit the impact of a hardware component failure. Minor component failures (e.g., defect hard drive) may not have a significant effect on the overall network. However, if they affect critical components (e.g., a backbone router) they could still cause availability issues in large parts of the private 5G network. Fewer technology/engineering resources dedicated to maintaining the network infrastructure (compared to a public MNO) can lead to greater recovery times as well. Therefore, the probability of these incidents is considered MEDIUM .	Medium
Design and architecture flaws	Availability (AP7-A)	Various design decisions in the infrastructure subsystem can affect the availability/resiliency of the overall network. Redundancy in power supply, network infrastructure, and physical compute/storage infrastructure should be a key consideration when planning any 5G network deployment, even smaller, non-public deployments. While private 5G network operators may not have access to immense resources to ensure highly available, geographically redundant deployments, the limited size of their 5G environments also makes it easier to ensure basic principles of resilient network design on a small scale. Therefore, the probability of severe design and architecture flaws that would result in a complete network failure is considered LOW .	Low

5.4.2 Degraded Services / Connections

The infrastructure subsystem provides the essential compute, storage, and networking resources to the workloads deployed on top of it. As such, the availability of the 5G System critically depends on the availability of this subsystem. Minor issues can also lead to a negative impact on the quality of 5G services.

Therefore, the **Infrastructure subsystem is considered** as a relevant attack scenario for the risk scenario Degraded Services / Connections.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	Misconfigurations of the Infrastructure subsystem can have a negative effect on the 3GPP service layer, such as limited resource allocation, flawed QoS, network congestion. In contrast to some grave misconfigurations at the 3GPP service layer that may entirely prevent connection establishment, some of these flaws may easily go unnoticed. Considering that the resources available for 5G infrastructure configuration management (incl. creating, testing, auditing configuration settings) will be limited and that the network's backbone is assumed to be part of the organization's general IT network (i.e., the 5G network may also be subject to misconfigurations of other network services), the probability of such incidents is considered HIGH .	High
Insufficient Access Controls	Availability (AP2-A)	Access control to the Infrastructure subsystem comprises logical access (e.g., to management or NFVI components) and physical access (i.e. to data centers). A gap in either of these may lead to a large-scale failure of availability. Assignment of overly broad logical access rights to internal staff can increase the likelihood of an incident, e.g., when a network administrator also has access to the platform on which Network Functions are hosted. This includes the timely revocation of access rights to prevent current or former employees from intentional acts of sabotage [47]. Since especially smaller organizations operating a private 5G network may not always have sufficient resources to enforce strict segregation of roles, necessitating the assignment of broader administrative rights, it may render them more likely to be the victim of internal attackers trying to compromise network services. Therefore, the probability of these incidents is considered MEDIUM .	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws in the Infrastructure subsystem (e.g., Hardware, Operating System, Hypervisor) may lead to negative effects at 3GPP service layer if they impact the performance of the software or virtualized workloads running on top of it. While not all implementation flaws result in a complete network outage [48], they may still cause a negative effect on service quality or availability. Ideally, severe issues should be caught by the testing processes of the technology provider. However, some private 5G operators may rely on smaller, less experienced vendors with less mature testing and quality assurance processes, making them more likely to be effected by such issues. Therefore, the probability of these incidents is considered HIGH .	High

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Specification Weakness	Availability (AP4-A)	Availability failures are generally not a direct result of a specification weakness. For ensuring availability, design, implementation, and test are much more relevant (see “Design and architecture flaws” below). Therefore, the probability of these incidents is considered LOW .	Low
Failure of Power Supply	Availability (AP5-A)	Smaller, private 5G deployments may not have the resources or expertise to maintain a highly redundant network deployment. Even if they do, the chance of a power supply failure at a company data center hosting central network components (e.g., 5G Core) affecting service quality to some extent is HIGH . Fewer technology/engineering resources dedicated to maintaining the network infrastructure (compared to a public MNO) can also lead to greater recovery times.	High
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	Similar to the case of a “Failure of Power Supply”, private deployments are expected to feature less redundancy and failover capacity that would limit the impact of a hardware component failure. Minor component failures (e.g., defect hard drive) may not have a significant effect on the overall network. However, if they affect critical components (e.g., CPU of a server hosting 5G Core NFs) they could still cause availability issues in large parts of the private 5G network. Moreover, fewer technology/engineering resources dedicated to maintaining the network infrastructure (compared to a public MNO) can lead to greater recovery times as well. Therefore, the probability of these incidents is considered MEDIUM	Medium
Design and architecture flaws	Availability (AP7-A)	Various design decisions in the infrastructure subsystem can affect the availability/resiliency of the overall network. While they may not lead to Degraded Services/Connections under normal circumstances, they may be the cause of sub-par performance in contingency situations (power outages, unexpected traffic spikes, etc.), making it harder to maintain the expected network performance or recover from an incident. Smaller organizations operating private 5G deployments may simply not have the resources and expertise to ensure high-availability under these circumstances while the expert support provided by their technology suppliers may also be limited. Therefore, the probability of such design and architecture shortcomings impacting service quality to some extent is considered MEDIUM .	Medium

5.4.3 Information Disclosure

This risk scenario focuses on failures in transport layer protection provided by the 5G System. While the Infrastructure subsystem hosts the 5G Network Functions providing the relevant security controls, it is not directly involved in ensuring secure data transmission according to 3GPP specification.

Therefore, **the Infrastructure subsystem is not considered** as a relevant attack domain for the risk scenario Information Disclosure.

5.4.4 User Identities (Authenticity/Integrity)

Verification of message authenticity/integrity based on user related identities (e.g., IMSI, IMEI, 5G-GUTI) or associated session information is not performed at infrastructure layer. Instead, such verification is performed at the 3GPP layer by the 5G Network Functions deployed on top.

Therefore, the **Infrastructure subsystem is not considered** as a relevant attack domain for the risk scenario User Identities.

6 Risk Assessment Results

This chapter presents the final results of the detailed risk analysis. The final risk analysis that covers all subsystems is presented in the framework document.

Note: Risk scenarios with no given probability level were not rated, see beginning of Chapter 5 and the corresponding risk scenario.

6.1 Probability Assessment

The following table summarizes the results of this detailed analysis of the **Infrastructure** layer in a 5G network. According to the maximum principle in risk analysis only the highest probability is listed.

Perspective	Section	Risk Scenario	Probability Level
Public MNO	5.2.1	Emergency Call (TKG § 164)	
	5.2.2	Cell Broadcast (TKG § 164a)	
	5.2.3	Telecommunications Secrecy (TTDSG § 3)	High
	5.2.4	Emergency Preparedness (TKG § 184)	
	5.2.5	Regional Network Failure	
	5.2.6	Multi-Regional Network Failure	Medium
	5.2.7	Failure of Mobile Service	
	5.2.8	Fraud	
	5.2.9	Subscriber-Data Leak	High
User	5.3.1	Urgent Telephone Calls Not Possible	
	5.3.2	Authenticity of the Receiver of a Call	
	5.3.3	Integrity of Transmitted Data	Medium
	5.3.4	User's Identity	High
	5.3.5	Eavesdropping of a Communication	Medium
	5.3.6	Slowing Down/Restricting the Connection	
	5.3.7	Connection Failure	
	5.3.8	Tapping or Altering of the Geographical Position	
Private 5G Network Operator	5.4.1	Network Failure	Medium
	5.4.2	Degraded Services/Connections	High
	5.4.3	Information Disclosure	
	5.4.4	User Identities (Authenticity/ Integrity)	

6.2 Risk Assessment

The following table summarizes the overall risk assessment of the **Infrastructure** layer in a 5G network. The risk scenarios and their respective impact level are described in the framework document.

Perspective	Risk Scenario	Impact	Probability	Risk Score
Public MNO	Emergency Call (TKG § 164)			
	Cell Broadcast (TKG § 164a)			
	Telecommunications Secrecy (TTDSG § 3)	Medium	High	High
	Emergency Preparedness (TKG § 184)			
	Regional Network Failure			
	Multi-Regional Network Failure	High	Medium	High
	Failure of Mobile Service			
	Fraud			
	Subscriber-Data Leak	Medium	High	High
User	Urgent Telephone Calls Not Possible			
	Authenticity of the Receiver of a Call			
	Integrity of Transmitted Data	High	Medium	High
	User's Identity	High	High	High
	Eavesdropping of a Communication	Medium	Medium	Medium
	Slowing Down/Restricting the Connection			
	Connection Failure			
	Tapping or Altering of the Geographical Position			
Private 5G Network Operator	Network Failure	Medium	Medium	Medium
	Degraded Services/Connections	Medium	High	High
	Information Disclosure			
	User Identities (Authenticity/ Integrity)			

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